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Accurate and efficient integrative reference-informed spatial domain detection for spatial transcriptomics

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Spatially resolved transcriptomics (SRT) studies are becoming increasingly common and large, offering unprecedented opportunities in mapping complex tissue structures and functions. Here we present integrative and reference-informed tissue segmentation (IRIS), a computational method designed to characterize tissue spatial organization in SRT studies through accurately and efficiently detecting spatial domains. IRIS uniquely leverages single-cell RNA sequencing data for reference-informed detection of biologically interpretable spatial domains, integrating multiple SRT slices while explicitly considering correlations both within and across slices. We demonstrate the advantages of IRIS through in-depth analysis of six SRT datasets encompassing diverse technologies, tissues, species and resolutions. In these applications, IRIS achieves substantial accuracy gains (39–1,083%) and speed improvements (4.6–666.0) in moderate-sized datasets, while representing the only method applicable for large datasets including Stereo-seq and 10x Xenium. As a result, IRIS reveals intricate brain structures, uncovers tumor microenvironment heterogeneity and detects structural changes in diabetes-affected testis, all with exceptional speed and accuracy.

Spatially resolved transcriptomics (SRT) are a set of recently developed technologies that enable gene expression profiling in tissues with spatial context. These SRT technologies include both imaging-based approaches, such as single molecular in situ hybridization (for example, MERFISH[1] and 10x Xenium[2]) or in situ sequencing (for example, STARmap[3]) and approaches based on next-generation sequencing (for example, spatial transcriptomics (ST)[4], 10x Visium[5], Slide-seq[6,7] and Stereo-seq[8], to name a few). These technologies have provided unprecedented opportunities for investigating and characterizing the transcriptomic and cellular landscape of complex tissues.

A major analytic task of SRT studies is to characterize the spatial organization of complex tissues through spatial domain detection[9,10]. Tissues are complex cellular ecosystems with spatially organized and functionally distinct anatomical domains and microenvironments, each characterized by unique cell type compositions and transcriptomic heterogeneity. The spatial organization of tissues in the form of local domains facilitates how different cell types coordinate with each other in carrying out tissue functions in development, homeostasis, communication, repair and signaling responses. Consequently, detecting spatial domains on the tissue in SRT studies can facilitate our understanding of the spatial and functional organization of a normal tissue and reveal how alterations in the tissue structure may underlie disease etiology. Recent computational advancements for spatial domain detection in SRT include the graph convolutional network method spaGCN[11], the Potts model-based methods such as

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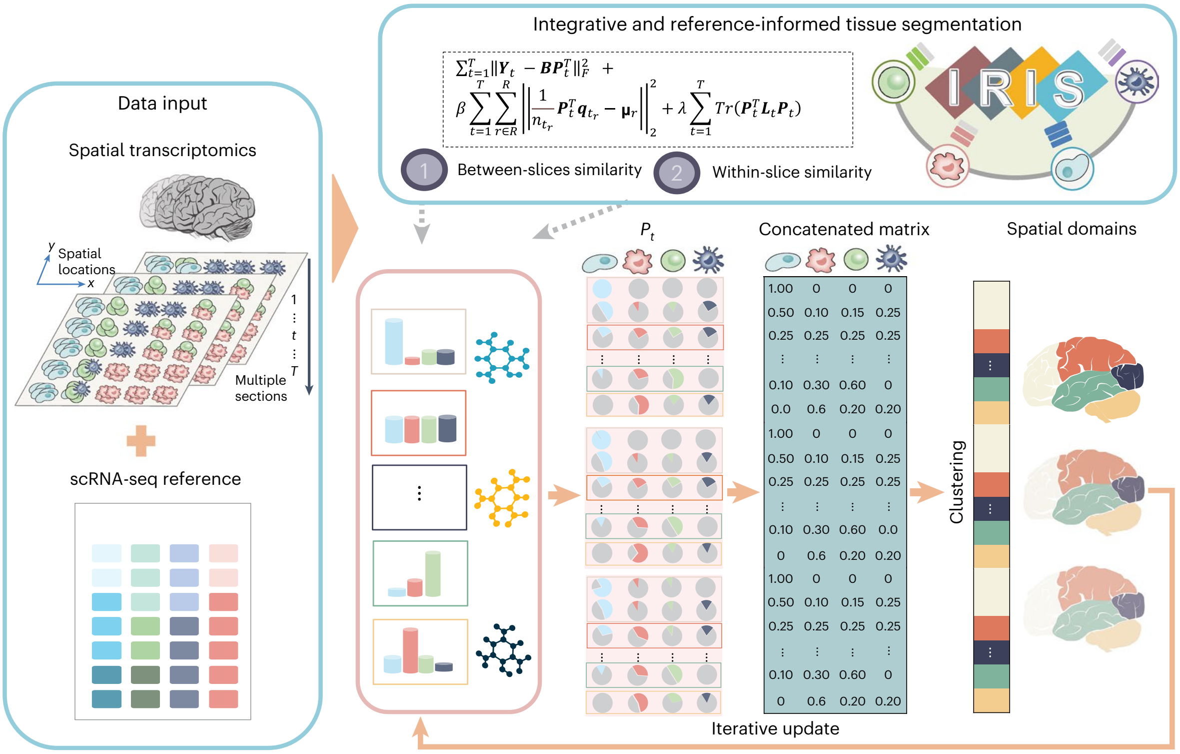

Fig. 1 | Schematic overview of IRIS. IRIS is an accurate and efficient integrative reference-informed segmentation method for detecting spatial domains on multiple tissue slices across a range of SRT technologies. IRIS requires as input SRT data measured on multiple tissue slices with spatial localization information, along with scRNA-seq reference data measured on the same tissue with cell-type-specific gene expression information. With these two data inputs, IRIS performs integrative and reference-informed domain detection by integrating scRNA-seq data into the SRT data and further integrating SRT data across multiple tissue slices. Such integrative analysis of IRIS is carried out through a joint modeling framework, with an efficient iterative optimization algorithm for ensuring scalable computation. One unique feature of IRIS is its ability to leverage cell-type-specific expression profiles from the reference scRNA-seq data to facilitate the mapping of spatial domains in SRT studies. Another unique feature of IRIS is its ability to incorporate the spatial transcriptomic profiles from neighboring spatial locations on each single tissue slice as well as that across multiple tissue slices, while properly accounting for both within- and between-slice correlations, to achieve accurate and consistent spatial domain detection across slices. As a result, IRIS is highly accurate, robust and scalable.

BayesSpace[12] and BASS[13], the autoencoder-based method SEDR[14], the graph self-supervised contrastive learning method GraphST[15], the graph attention autoencoder STAGATE[16], and the hybrid deep learning and Bayesian framework Maple[17], to name a few. Unfortunately, almost all existing methods directly rely on transcriptomic heterogeneity, which is a secondary characteristic resulting from the primary feature of unique cell type composition underlying each spatial domain, to disentangle tissue structure. However, modeling the secondary feature of transcriptomic heterogeneity instead of the primarily feature of cell type composition for spatial domain detection is not ideal, as such an approach makes it difficult to characterize the cellular landscape of the tissue, reduces the accuracy and interpretability of the detected spatial domains and, as will be shown here, often leads to the identification of biologically irrelevant structures.

In this Article, we present an alternative strategy for detecting spatial domains in SRT studies. Specifically, we directly model the primary feature of cell type compositional heterogeneity across spatial locations and use it to segment the tissue into multiple biologically relevant spatial domains, each characterized by a distinct cell type composition. This alternative strategy has four important benefits. First, it directly characterizes the cellular landscape of the tissue and identifies biologically interpretable spatial domains, thus facilitating the understanding of the cellular mechanism underlying tissue function. Second, it offers a framework for integrating cell-type-specific transcriptomic profiles[18,19] obtained from readily available single-cell RNA sequencing (scRNA-seq) data into SRT. This enables us to leverage external scRNA-seq datasets, which are almost always available for the tissue samples used in the SRT study, leading to potentially substantial accuracy gain for spatial domain detection. Third, by focusing on the primary feature of cell type composition, the alternative strategy naturally provides an anchoring point for integrating SRT data across multiple tissue slices or samples that are commonly collected in recent SRT studies but not yet analyzable by most existing methods. In particular, because multiple slices from the same tissues often contain similar spatial domains characterized by similar cell type compositions, anchoring them on the shared domain-specific cell type composition allows borrowing similarity information from multiple slices to enhance spatial domain detection. Finally, our strategy transforms spatial domain detection from focusing on transcriptomic heterogeneity to a conceptually simpler task of characterizing domains on the basis of domain-specific cell type compositions. This shift paves ways for various computationally efficient algorithms, making the detection of spatial domains feasible in very large-scale SRT datasets that are currently formidable to almost all existing domain detection methods.

While the above alternative strategy is biologically intuitive, effective implementing it, however, is nontrivial. As we show below, a naive application of this alternative strategy, which first infers cell type compositions on the tissue and then conducts tissue segmentation via clustering on the inferred compositions, does not produce spatial domains as accurate as the current methods. Consequently, to harness this alternative strategy's full potential, we have developed a novel computational method, integrative and reference-informed tissue segmentation (IRIS), for spatial domain detection. IRIS is accurate, scalable and robust, showcasing its advantages across six SRT datasets from different tissues and species, covering 37 tissue slices sequenced using 10x Visium, ST, Slide-seq, Vizgen MERFISH, Stereo-seq and 10x Xenium. Our results show that IRIS considerably surpassed existing methods in accuracy for spatial domain detection with substantially greater computational efficiency. The unparalleled accuracy and computational gains delivered by IRIS make it an indispensable tool for integrated tissue segmentation in SRT studies.

Results

Method overview

IRIS is described in Methods, with its technical details provided in Supplementary Note 1 and its method schematic shown in Fig. 1. Briefly, IRIS is a reference-informed integrative method for detecting spatial domains on multiple tissue slices from SRT with spot-level, single-cell level or subcellular-level resolutions. IRIS builds upon the fact that each spatial domain on the tissue is characterized by its unique cell type composition and that similar composition is observed for the same spatial domain across different slices of the same tissue type[18,20]. Consequently, IRIS integrates a reference scRNA-seq data to inform and characterize the cell type composition on the SRT tissue for accurate and interpretable spatial domain detection. In the process, IRIS accommodates cell type compositional similarity across locations within the same slice and across different slices on the same domain. Importantly, IRIS comes with an efficient optimization framework with multiple algebraic innovations for scalable computation and can easily handle multiple SRT datasets with millions of spatial locations and tens of thousands of genes.

Human DLPFC 10x Visium data

To benchmark IRIS against other spatial domain detection methods, we first examined the gold-standard human dorsolateral prefrontal cortex (DLPFC) data by 10x Visium, featuring 12 slices from three donors, with annotations for seven spatial domains including six cortical layers and white matter[21]. We obtained scRNA-seq data from 10x Chromium on the post-mortem brain tissue with 44 cell types as reference[22] (Supplementary Table 1). Note that the scRNA-seq reference is from an external study with samples consisting of cell types and states that are potentially different from those in SRT. We used domain annotations from ref. 21 as the ground truth, evaluated spatial domain detection accuracy by the adjusted Rand index (ARI)[11,12] (Supplementary Note 2) and compared IRIS with multiple state-of-the-art domain detection methods (Methods).

We first examined a simple setting where tissue slices come from the same donor, with similar tissue structures shared across slices (Supplementary Table 2). In the analysis, IRIS correctly detects the layered structures of the prefrontal cortex (median ARI 0.71; Fig. 2a), representing 39–1,083% accuracy gain compared with the other methods (Fig. 2b, Supplementary Note 3 and Supplementary Fig. 1). In addition, IRIS clearly captures the sandwich-like structure of the cortical layers 1–3 that is matched with annotation and detects the deep layers 4–6 where the other methods cannot distinguish well. Regardless of the prespecified number of spatial domains, IRIS always performs the best and the comparative results remain consistent (Fig. 2c).

Next, we evaluated more challenging settings. First, we examined settings with missing or misclassified cell types (Supplementary Note 4) in the scRNA-seq reference. Here, IRIS remains superior (median ARI 0.66) with a 29–1,000% accuracy gain over other methods, for missing (Supplementary Figs. 2 and 3) or misclassified cell types (Supplementary Figs. 2 and 4). Second, we used an alternative scRNA-seq reference[23] with distinct cell types and found that IRIS maintains high consistency (ARI for consistency 0.70 to 0.86) and outperforms other methods (Supplementary Figs. 5 and 6). Third, we analyzed tissue slices from different donors with distinct tissue structures (Supplementary Table 2) and found IRIS to be accurate again (median ARI 0.71), outperforming other methods by 20–1,083% (Supplementary Figs. 7 and 8), with the relative performance remaining consistent in both baseline and more complex settings (Fig. 2c and Supplementary Figs. 2–8). Fourth, IRIS maintained its accuracy across various sequencing downsampling scenarios (Supplementary Note 4 and Supplementary Fig. 9). Finally, careful examination of the first and third settings revealed that IRIS's high accuracy results from effective incorporation of scRNA-seq reference data when SRT tissue slices are homogeneous and results from integrating both scRNA-seq reference and multislice modeling when SRT slices are heterogeneous (Supplementary Figs. 10 and 11, and Supplementary Note 5).

To validate IRIS-detected spatial domains, we first examined the layer specific marker genes (Methods) and found that the identified layers by IRIS are enriched with known marker genes (Fig. 2d and Supplementary Table 3). Second, we performed downstream analysis to characterize the transcriptomic and cellular landscape of tissue domains (Methods). Differential expression (DE) analysis identified genes that are specifically expressed in different domains, including both known (that is, *CARTPT*[21] and *PCP4* (ref. 24)) and novel marker genes (that is, *APOE*[25], *NRGN* and *PLP1* (ref. 26)) (Supplementary Fig. 12 and Supplementary Table 4). For example, *APOE* is an identified DE gene in spatial domain 0 (layer 1) and encodes apolipoprotein E, which is central for lipid metabolism[25]. *PLP1* is an identified DE gene in spatial domain 6 (white matter) and is closely related to myelination occurring there[26]. Further gene set enrichment analysis (GSEA) on DE genes revealed enrichment in pathways of synapse signaling, post-synapse signaling and Alzheimer disease, all of which are hallmarks of brain functionality (Fig. 2e, Supplementary Fig. 13 and Supplementary Note 6). Third, examining cell type compositions uncovered a nuanced colocalization of various cell types within specific domains, such as a mixture of astrocyte subtypes, oligodendrocyte progenitor subtypes and pericytes in domain 0 (layer 1), different excitatory neuron subtypes across domains 1–4 (cortical layers 2–5) and various oligodendrocyte subtypes enriched in domain 6 (white matter) and inhibitory neuronal subtypes enriched in domains 3–4 (deeper layers 4–5)[27]. Similar results are observed by examining the spatial distribution of the representative cell types (Fig. 2g) and are apparent across tissue slices (Supplementary Figs. 14 and 16, and Supplementary Table 2).

Fig. 2 | Analyzing the human DLPFC 10x Visium data. a, Spatial domains detected by IRIS and other methods. Results are shown in the baseline analysis setting where the tissue slices are from the same donor (denoted as sameStr). Ground-truth tissue regions of the human prefrontal cortex are obtained from the original DLPFC study (left). Clustering accuracy of different methods is measured by ARI, with a higher ARI indicating higher accuracy. **b**, Boxplots displaying the clustering accuracy measured in the format of ARI on the tissue slices ($n = 4$) applied in the baseline setting (sameStr). Each box plot ranges from the first to third quartiles with the median as the horizontal line, while whiskers represent 1.5 times the interquartile range from the lower and upper bounds of the box. **c**, The clustering performance of different methods when varying the prespecified number of spatial domains. The median ARI across all slices in either setting was calculated to measure the spatial domain detection accuracy. **d**, Top: lollipop plots displaying the mean expression of seven important marker genes (y axis) in each spatial domain identified by IRIS (x axis). The solid arrow indicates the highest enrichment value. Bottom: bar plots displaying the fold change of each marker gene in the expected domain versus other domains, where a higher value indicates higher domain detection accuracy. **e**, The top enriched gene sets in the selected spatial domains detected by IRIS in GSEA, with P values calculated on the basis of an adaptive multilevel split Monte-Carlo scheme implemented in the fgsea package[77]. **f**, A heat map plot displaying the estimated mean cell type proportion for each cell type in each spatial domain detected by IRIS. The color scale was normalized to the 0–1 range. **g**, A spatial scatter plot displaying the spatial distribution of the IRIS-estimated cell type proportion for each cell type across spatial locations. For **d**–**g**, the results are shown for the example slice 151509 in the sameStr analysis. Ast, astrocytes; Ex, excitatory neurons; End, endothelial cells; Oli, oligodendrocytes; Opc, oligodendrocyte precursor cell; Per, pericytes; In, inhibitory neurons; WM, white matter.

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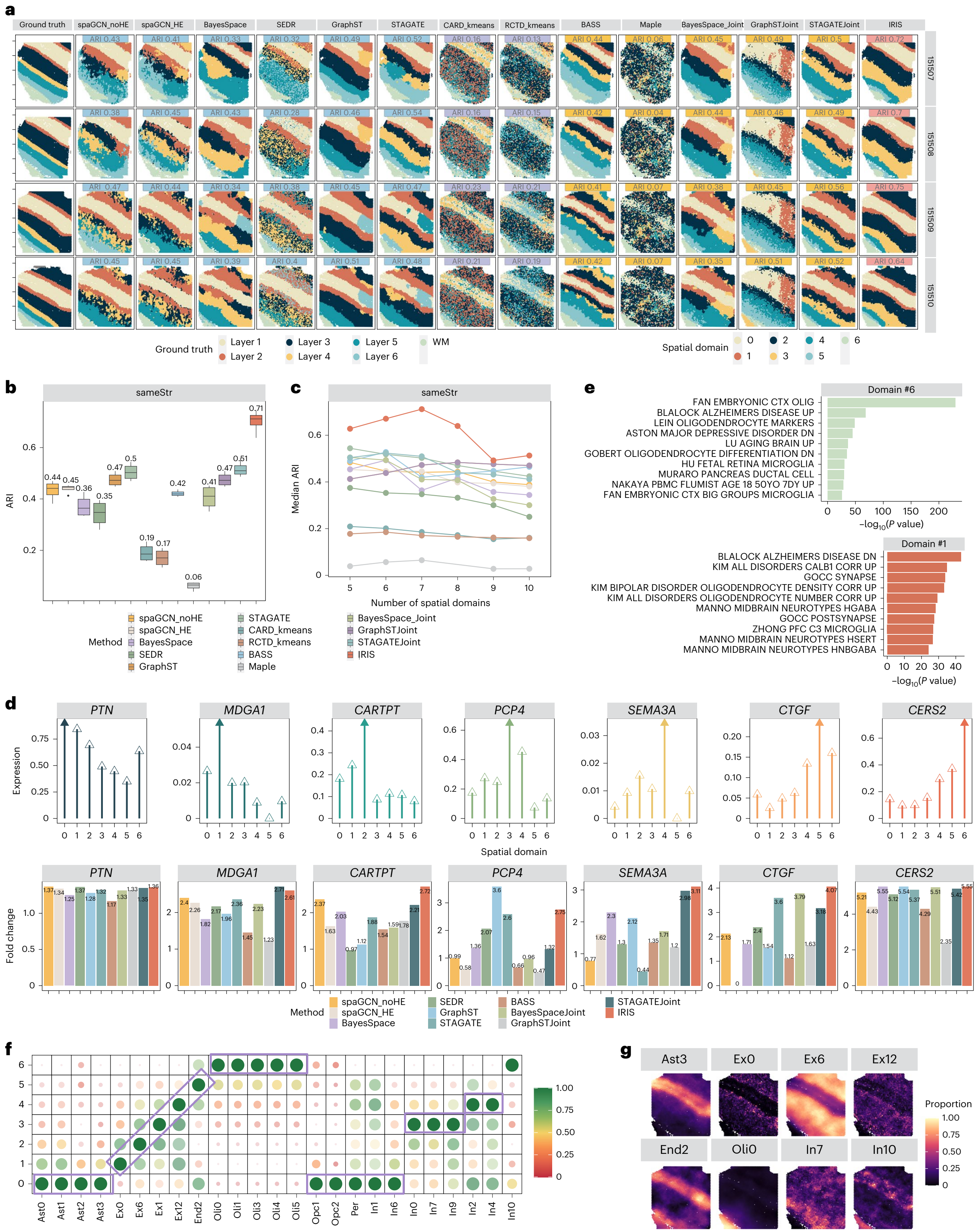
Human SCC ST data

The second data we examined is human squamous cell carcinoma (SCC) data from ST, consisting of 12 slices across four patients. We followed ref. 28 to focus on three tissue slices from patient 2 as the main example. SCC is the second most common form of skin cancer caused by malignant proliferation of keratinocytes[28], and the characterization



of tumor–immune dynamics in the tumor microenvironment (TME) is crucial for its prognosis. For scRNA-seq reference, we used data from the same study, which consists of 24 cell types (Supplementary Table 1).

The SCC data do not contain detailed domain annotations applicable for method evaluation. Thus, we used the tumor versus nontumor regions annotated by histologists on the basis of the tumor leading edges on the tissue slice (P2_ST_rep2) as the silver standard for coarse but not refined domain segregation evaluation (Fig. 3a and Supplementary Fig. 17). IRIS clearly captures the coarse regional segregation between tumor and nontumor regions (Methods, Nagelkerke $R^2 = 0.72$), more so than the other methods (Fig. 3b,c, Nagelkerke $R^2 = 0.4–0.71$ for other methods). In addition, perhaps more importantly, IRIS can further identify the tumor core regions in the center of the tumor (for example, detected domains 3 and 6) and separate them from the leading edges of the tumor (for example, domain 8) across slices while the other methods failed to clearly do so (Fig. 3b,d, Supplementary Fig. 18 and Supplementary Note 7). For the nontumor regions, IRIS consistently detected multiple domains indicative of TME, including the tumor-adjacent (domain 4) and nonadjacent stroma regions (domains 2, 5, 9 and 7), while the other methods did not capture these regions consistently across slices. Spatial continuity and compactness analysis quantified by the CHAOS score (Supplementary Note 2) revealed that IRIS, STAGATEJoint and GraphSTJoint had slightly better spatial continuity and compactness, though the latter two methods misidentified spatial domains including both tumor cores and edges (Fig. 3b and Supplementary Fig. 19). The spatial continuity results are consistent regardless of the prespecified number of spatial domains and which patient samples were used (Supplementary Fig. 19).

Next, we validate IRIS's performance through examining marker gene expression in the detected spatial domains (Methods). We found that only IRIS was able to separate the tumor region into three subregions (6, 3 and 8) in an inside-out fashion, while the other methods either failed to detect the tumor edge (domain 8) or separate the tumor region into two subregions 6 and 3. The known tumor core[28] and tumor edge marker genes[28] (Supplementary Table 5) showed 8–72% stronger enrichment in IRIS-detected domains compared with the other methods that were able to detect the tumor core and edge regions (Fig. 3e). Domain 6, identified as the tumor core, appears to be a well-differentiated SCC region as the differentiating keratinocytes are highly enriched there (Fig. 3f,g and Supplementary Fig. 20), while domain 8, the tumor edge, showed a high abundance of cycling cancer cells[29] and immune cells (for example, dendric cell subsets including CLEC9A, LC, ASDC and CD1C). The novel domain 3 is statistically significantly near both domains 6 and 8 in the nearest-neighbor analysis (Methods and Fig. 3h) but is not enriched by the aforementioned tumor core or edge marker genes, representing a unique SCC structural component (Supplementary Fig. 21). DE analysis comparing domain 3 with domain 6 identified multiple upregulated genes related to keratinocyte differentiation and cell adhesion (that is, *KRT14*, *LY6D*, *DSG1*, *S100A16* and *DSC3*) and downregulated genes (that is, *VIM*, *MMP1* and *COL1A1*) belonging to the epithelial–mesenchymal transition pathway in domain 6 (Fig. 3i and Supplementary Table 6), suggesting that domain 3 is a less differentiated but more malignant tumor region with increased metastatic activity[30]. DE analysis comparing domain 3 with domain 8 identified multiple upregulated genes (that is, *NFATC1* and *HLA.C*) related to infiltrating immune systems[31,32] in domain 8, supporting an immune interface at the tumor edge (Fig. 3i). GSEA further confirmed these findings, with enriched gene sets related to suprabasal cell signature and immune pathways (Supplementary Fig. 22). The results by IRIS in the tumor region highlight the heterogeneity of SCC and the transcriptomic and cellular transition from the tumor core toward the tumor edge. Consistently, IRIS identified distinct stroma and immune-related domains in nontumor regions, revealing spatial and cellular heterogeneity with varied enrichment of tumor-suppressor genes (for example, *S100A2* and *COL7A1*) and immune cell subsets across tumor-adjacent and nontumor-adjacent stroma (Fig. 3f, Supplementary Fig. 23, Supplementary Table 7 and Supplementary Note 7). Similar results are observed in the other three tissue slices (Supplementary Figs. 24 and 25).

Mouse spermatogenesis Slide-seq data

Next, we analyzed the mouse spermatogenesis Slide-seq data[33] collected on testis, consisting of well-defined structures in seminiferous tubes that can be easily visualized to evaluate method performance. This study sequenced six tissue slices from three diabetic (ob/ob) and three wild-type (WT) mice. The scRNA-seq reference was obtained from an external study by Drop-seq technology[34] that measured gene expression on six adult mice. We primarily focused on one WT slice and one ob/ob slice, to investigate diabetes-induced structural changes (Supplementary Table 2). For this and the following datasets, we excluded deconvolution-based methods due to their poor performance and we were unable to apply GraphST and Maple due to their heavy computational burden.

Spermatogenesis involves five successive stages: spermatogonia (SPG), primary spermatocytes (SPC), secondary SPC, round spermatids (RS), elongating spermatids (ES) and spermatozoa[35], all organized spatially within the seminiferous tubules (Fig. 4a). IRIS accurately depicted the expected structure of the seminiferous tubules (Fig. 4b). Specifically, in WT mouse (Supplementary Table 2), IRIS captured a spatial domain (domain 0) primarily located in the center of the seminiferous tubules where RS resides; a domain (domain 2) in the peripheral where ES resides; two domains (domains 3 and 4) colocalized with domain 2, probably representing primary and secondary SPC; and a domain (domain 1) capturing the interstitial space with SPG, Leydig cells, Sertoli cells, endothelial cells and macrophage cells. In contrast, the spatial domains detected by spaGCN, BayesSpace and BayesSpaceJoint displayed a chaotic pattern not aligning with known seminiferous tubular structures, including RS and SPC. SEDR and BASS

Fig. 3 | Analyzing the human SCC ST data. a, An H&E staining of the human SCC tissue (P2_ST_rep2) displays the tumor leading edges that coarsely segregate the tumor regions from nontumor regions. **b**, Spatial domains detected by IRIS, and the other methods in the analysis of patient 2. **c**, Pseudo-R^2 measures the degree to which the spatial domains inferred by different methods can be used to differentiate the tumor regions from the other regions. Two types of pseudo-R^2 measurements were used. **d**, The three-tumor related spatial domains detected by IRIS were separately colored by red (domain 6), blue (domain 3) and black (domain 8). **e**, Box plots quantifying the performance of different methods in detecting the tumor core and tumor edge by comparing the marker genes' enrichment ($n = 5$ marker genes for tumor core, and $n = 11$ for tumor edge) in the form of fold change. Each box plot ranges from the first to third quartiles with the median as the horizontal line, while whiskers represent 1.5 times the interquartile range from the lower and upper bounds of the box. **f**, A heat map plot displaying the estimated mean cell type proportion for each cell type in each spatial domain detected by IRIS. The color scale was normalized to the 0–1 range. The cell types encased in purple rectangles are highlighted as examples in the main text to illustrate the unique cell type compositions for spatial domains 4–6 and 8. **g**, A spatial scatter plot displaying the spatial distribution of the IRIS-estimated cell type proportion for each cell type across spatial locations. **h**, A heat map showing the number of nearest-neighbor identities for each spatial domain, where the asterisk indicates $P < 0.001$ by permutation test (one-sided). **i**, Volcano plots displaying the pairwise DE analysis results between tumor core (domain 6), tumor edge (domain 8) and novel region (domain 3), with P values calculated by Wilcoxon rank sum test. FC, fold change. For **c**–**i**, the results are shown for the main example slice P2_ST_rep2 of patient 2. KC, keratinocyte; KC_Cyc, cycling keratinocyte; KC_Diff, differentiating keratinocyte; KC_Basal, basal keratinocyte; TSK, tumor-specific keratinocyte; Mac, macrophages; PDC, plasmacytoid dendritic cell; MDSC, myeloid-derived suppressor cell.

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captured the general structure of ES colocalized with SPC but failed to identify the circular-shaped RS domain and incorrectly detected a smaller than expected ES domain (Fig. 4b). STAGATE's spatial domains did not capture the general structure of RS (domain 0) in the ob/ob mouse, suggesting unstable performance across tissue slices. Similarly, STAGATEJoint did not identify the structures of ES (domain 2) and RS

(domain 0). The superior performance of IRIS also applies to the ob/ob mouse, where only IRIS accurately identified ES domain and capture its reduced size compared with WT mice, supporting the disrupted spatial cellular architecture of the seminiferous tubules in diabetic mice[33]. Quantification shows that the spatial domains detected by IRIS (median CHAOS 0.014) and BASS (0.014) display better spatial continuity and compactness than the other methods (Fig. 4c), regardless of the prespecified number of spatial domains and which slice combination (Supplementary Table 2) was used (Supplementary Fig. 26).

We examined the expression pattern of known spermatogenesis-related genes[33] to further validate and quantify the IRIS-identified spatial domains (Fig. 4d). The ES- and RS-related domains (domains 2 and 0) are enriched with corresponding marker genes including *Prm1* (ref. 36), *Tnp1* (ref. 37) for ES, and *Tekt2* (ref. 38) and *Tex36* (ref. 39) for RS. Two domains identified by IRIS, probably corresponding to the primary SPC (domain 3) and secondary SPC (domain 4), are enriched with marker genes *Mllt10* (https://www.proteinatlas.org/ENSG00000078403-MLLT10/tissue/testis) and *Aurka*[40], respectively (Supplementary Fig. 27 and Supplementary Note 8). Specifically, gene *Mllt10* was previously found to be enriched in early spermatids and pachytene/diplotene SPC and *Aurka* is important for the secondary SPC stage during meiosis[40]. Domain 1 was enriched with mitochondrial genes and Sertoli cell marker genes (for example, *mt-Rnr2* and *Clu*[41]), probably representing the germinal epithelium (Supplementary Fig. 28). Other examples are shown in Supplementary Figs. 29–32. Quantifications with marker gene enrichment also highlight IRIS's accuracy compared with other methods such as SEDR, and BASS, which also identified the general structure of seminiferous tubules (Fig. 4e).

We carefully examined the molecular and cellular signatures of the spatial domains detected by IRIS. First, domain-specific DE analysis identified a median of 4,077 DE genes across five domains, revealing both known (for example, *Tekt2*, *Tex36*, *Tnp1*, *Prm1*, *Aurka* and *Clu*) and novel DE genes (for example, *Kif2b*, *Sycp1*, *Gsg1*, *Lyar* and *Ldhc*; Supplementary Fig. 33, Supplementary Table 8 and detailed interpretation in Supplementary Note 8). Additionally, DE analysis revealed substantial transcriptomic differences between the visually similar domains 3 and 4 (Supplementary Fig. 34 and Supplementary Note 8). GSEA analysis further found that the domain-specific DE genes are highly enriched in pathways related to cilium cellular component, male gamete generation and cell cycle (Fig. 4f, Supplementary Fig. 35 and Supplementary Note 8). Next, we found that each domain is often characterized by domain-specific cell types (Fig. 4g,h), such as ES cells in domain 2, RS cells in domain 0, SPC in domains 3 and 4, and a cell mixture including Sertoli, Leydig and endothelial cells in domain 1, all discernible by their spatial distributions (Fig. 4h).

Importantly, IRIS revealed critical changes in the spatial organization of testicular microenvironment under diabetic conditions, notably in the ES region's structure and cellular composition. In WT mice, this region is highly concentrated in the center of seminiferous tubes, whereas in the ob/ob mice, it becomes diffused with increased intermingling with other cell types (Fig. 4b). Purity analysis (Methods) revealed reduced ES domain and cell type purity under diabetic conditions (Fig. 4k), suggesting a loss of ES/spermatozoon in the ob/ob testes[17,18,33,42]. This is further evidenced by the downregulation of ES marker genes such as *Prm1*, *Prm2*, *Odf1* and *Smcp* (Fig. 4i and Supplementary Fig. 36) by 30–59%, in expression levels after diabetes (Fig. 4j). Additionally, several mitochondrial genes (that is, *mt-Rnr1* and *mt-Rnr2*) are upregulated in domain 1 in the ob/ob mice, consistent with mitochondrial dysfunction in the pathogenesis of diabetes[43,44].

Mouse olfactory bulb Stereo-seq data

We applied IRIS to a single-cell resolution SRT dataset on adult mouse brain by Vizgen MERFISH (detailed results in Supplementary Note 9, Extended Data Fig. 1, Supplementary Figs. 37–52 and Supplementary Table 9), and subcellular-resolution SRT data collected on two adjacent mouse olfactory bulb tissue slices from Stereo-seq[8]. For the reference, we obtained the scRNA-seq sequenced through 10x Chromium, consisting of 18 cell subpopulations[45]. In these data, we were only able to apply IRIS as the other methods failed to run due to heavy computational burden.

IRIS accurately depicted the layered structure in the olfactory bulb, including the subependymal layer (SEL)[46], granule cell layer (GCL), internal plexiform layer (IPL), mitral cell layer (MCL), external plexiform layer (EPL), glomerular layer (GL), olfactory nerve layer (ONL) and meninges[47] (Fig. 5a,b and Supplementary Fig. 53) regardless of the specified number of spatial domains (Fig. 5c). IRIS outperforms the reference-free version of IRIS (Supplementary Fig. 54 and Supplementary Note 9) and remains consistent across various downsampling scenarios (Supplementary Fig. 55). The identified layered structure is supported by the enrichment of known marker genes (Fig. 5d,e), such as *Ptgds*[48] for meninge enriched in domain 1, *S100a5* (ref. 45) for ONL layer (domain 2), *Mgst1* (ref. 49) for olfactory ensheathing cell enriched in a peripheral sublayer of ONL (domain 3), and *Apold1* (ref. 50) for periglomerular cells (PGCs) highly concentrated in domain 6. IRIS also differentiates the outer/superficial (domain 4) and inner/deep (domain 10) layers of EPL as evidenced by distinct spatial patterns of *Cbln4* (ref. 46) and *Ly6g6e*[51]. Other examples include MCL interneuron marker *Vip*[52] (enriched in domain 7), retina IPL marker gene *Slc17a7* (domain 5), granule cell marker *Tpbg* (domain 9), new marker gene *Penk*[53] (domain 8) and immature neuron[54] marker *Sox11* (domain 0), with quantifications further confirming these patterns (Fig. 5e, Supplementary Note 11 and Supplementary Fig. 56).

We performed additional analysis to further examine the molecular and cellular signatures of IRIS-detected spatial domains. First, domain-specific DE analysis identified a median of 1,376 genes across 11 domains. The identified DE genes include known marker (for example, *Sox11*, *Ptgds*, *S100a5*, *Ly6g6e*, *Slc17a7*, *Apold1*, *Vip*, *Tpbg* and *Cbln4*) and novel DE genes (for example, *Fabp7* (ref. 55), *Meis2* (ref. 56) and *Cck*[57]; Supplementary Fig. 57 and Supplementary Table 10). For example, the identified ONL (domain 2) specific gene *Fabp7* is heavily expressed in the ensheathing glial cells of the olfactory nerve, promoting the

Fig. 4 | Analyzing the mouse spermatogenesis Slide-seq data. a, A diagram displaying the biology process underlying spermatogenesis, including SPG, primary and secondary SPC, RS and ES, and spermatozoa. **b**, Spatial domains detected by IRIS and the other methods in the 'WT_Diabetes' analysis. **c**, Box plots displaying CHAOS, which measured the spatial continuity and compactness of the detected spatial domains from different methods, on the tissue slices ($n = 2$) in the 'WT_Diabetes' analysis (left). The line plots display CHAOS values when varying the prespecified number of spatial domains, with the median CHAOSs used. **d**, Top: scatter plots displaying the spatial distribution of important spermatogenesis-related marker genes. Bottom: lollipop plots (bottom) displaying the mean expression of important marker genes (y axis) in each IRIS-identified spatial domain (x axis). The solid arrow indicates the highest enrichment value. **e**, Bar plots displaying the fold change of the marker gene expression in the expected domain versus other domains, where a higher value indicates better spatial domain detection accuracy. **f**, Top enriched gene sets in selected spatial domains (for example, domains 0 and 2), with P values calculated on the basis of an adaptive multilevel split Monte-Carlo scheme implemented in the fgsea package[77]. **g**, A heatmap plot displaying the estimated mean cell type proportion for each cell type in each spatial domain detected by IRIS. The color scale was normalized to the 0–1 range. **h**, A spatial scatter plot displaying the spatial distribution of the IRIS-estimated cell type proportion for each cell type across spatial locations. **i**, A comparison of the spatial pattern of ES marker genes in the WT mice and ob/ob (diabetic) mice. **j**, A bar plot displaying the mean expression of ES marker genes in the ES domain (domain 2; the first four panels) and the mean expression of mitochondrial genes in domain 1 (the latter two images) detected by IRIS in WT (olive) and ob/ob (pink) slices. **k**, The purity score of the ES domain (top) and ES cells (bottom) in the WT (blue) and ob/ob (yellow) mice. For **d**–**g**, the results are shown for the example WT slice (WT3_Puck7) in the main analysis.

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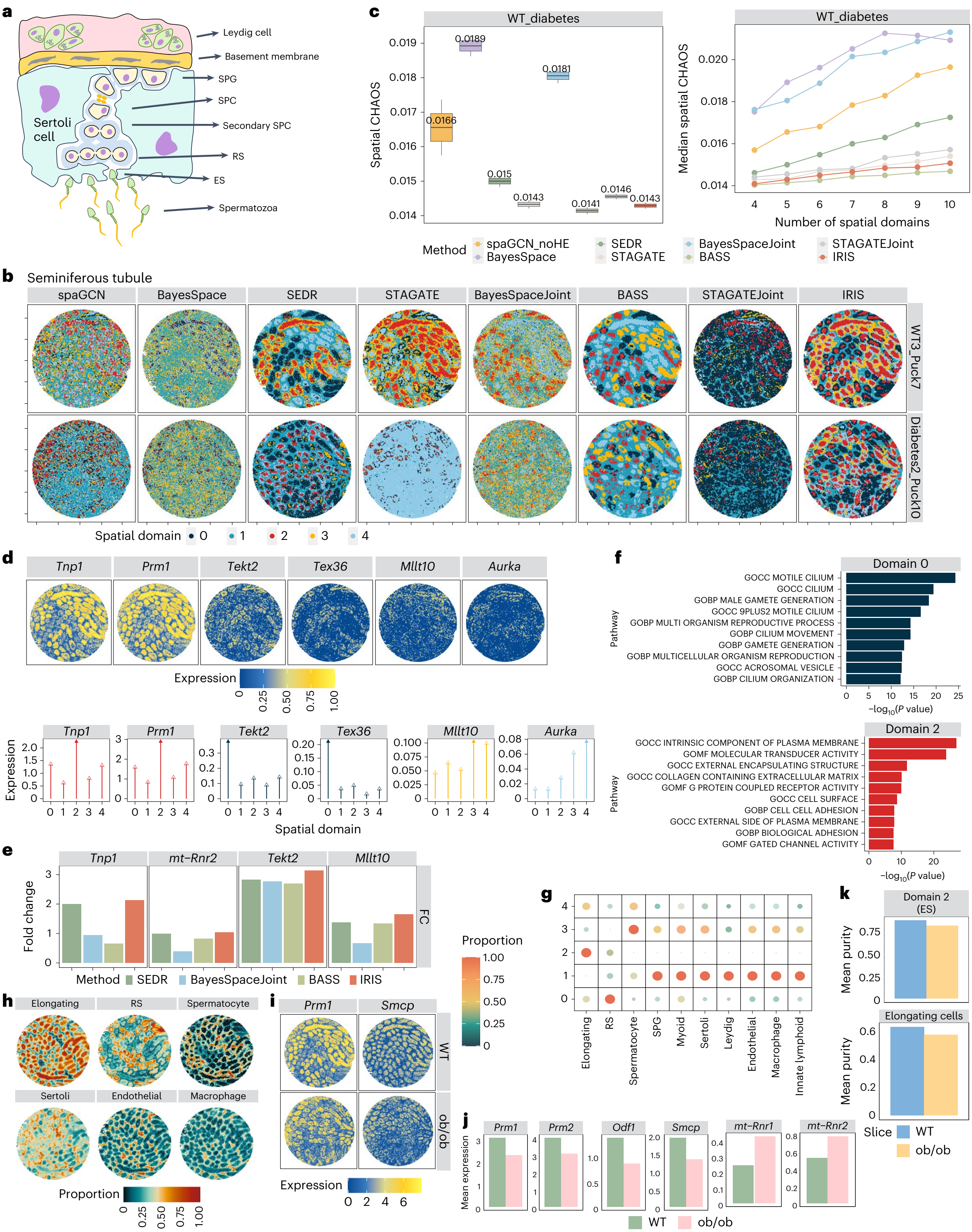
establishment and regenerative growth in sensory neurons in ONL[55]. The identified inner/deep EPL (domain 10) specific DE gene *Cck* is expressed preferentially in tufted cells mainly located in deep EPL[57]. GSEA further revealed enriched pathways related to olfactory receptor activity, sensory perception of smell, and odorant binding (Fig. 5f and Supplementary Fig. 58). The examination of IRIS-inferred cell type compositions



further highlighted unique characteristics within these domains, such as mitral cell subtypes across adjacent layers (4, 10, 5 and 7), granule cell subpopulations within distinct GCL sublayers (8 and 9) (Fig. 5g,h and Supplementary Fig. 59) and immature cells within domain 0. Similar results were observed in the other replicate (Supplementary Fig. 60).

High-resolution human BC 10x Xenium data

Finally, we applied IRIS to a high-resolution SRT dataset generated by 10x Xenium[2] on two adjacent human breast cancer (BC) tissue slices. We obtained the scRNA-seq data sequenced through 10x Chromium, consisting of 29 cell subpopulations[58]. We also explored the use of alternative scRNA-seq data that yielded consistent results (Supplementary Fig. 61 and Supplementary Note 12). In these data, again we were only able to apply IRIS as all the other methods failed to run due to heavy computational burden.

BC is heterogeneous with high intratumoral and intertumoral variation in histological and molecular features. In these data, IRIS clearly identified four distinct tumor domains, including two ductal carcinoma in situ (DCIS) domains that represent noninvasive forms of BC (domains 10 and 16) and two invasive ductal carcinoma (IDC) domains (domains 5 and 8) (Fig. 6a and Supplementary Fig. 62), along with multiple other domains that belong to part of the TME, including the immune-related regions (domains 0, 4 and 13), tumor stroma region (domain 1) and myoepithelial layer (domain 14). IRIS-detected spatial domains are spatially smooth and compact, regardless of the prespecified number of spatial domains (Fig. 6b), moreso than the reference free version of IRIS (Supplementary Figs. 63 and Supplementary Note 10). These spatial domains are further supported by the enrichment of known marker genes (Fig. 6c,d). For instance, *TCIM* (*TC-1*), a candidate BC oncogene associated with β-catenin pathway that elevates the invasive behavior of cancer cells[59], is highly enriched in the IDC domains 5 and 8. *FASN*[2], an invasive tumor marker gene, is highly enriched in the IDC domain 8. *CEACAM6*, which acts as a cell adhesion molecular with expression negatively correlated with cell differentiation[2,60], is highly enriched in the DCIS domains 10 and 16. Besides the tumor domains, IRIS distinguishes multiple immune-related domains characterized by their unique cellular compositions and different marker genes (Fig. 6d and Supplementary Note 12).

We performed additional analysis to examine the molecular and cellular signatures of the spatial domains detected by IRIS. First, domain-specific DE analysis and identified a median of 141 genes across 17 domains. The identified DE genes include known marker (for example, *TCIM*, *FASN*, *CEACAM6*, *FGL2*, *BANK1*, *IL7R*, *POSTN* and *KRT6B*) and novel DE genes (for example, *NDUFA4L2*, *ABCC11*, *EPCAM* and *CXCR4*; Supplementary Fig. 64 and Supplementary Table 11, with a detailed explanation in Supplementary Note 12). GSEA further indicated enrichment in specific pathways across tumor, immune and stromal domains, reflecting the complex interplay within the BC microenvironment (Fig. 6e, Supplementary Fig. 65 and Supplementary Note 12). Next, an examination of IRIS inferred cell type compositions found that the spatial domains identified by IRIS consist of unique cell type characteristics (Fig. 6f and Supplementary Fig. 66), with cancer cell subpopulations (cycling, Her2, luminal like and basal) highly enriched in tumor domains 5, 8, 10 and 16, with the latter two domains also enriched with myoepithelial, luminal progenitors and mature luminal cells. The majority of immune cells (for example, B cells, T cells, natural killer (NK) cells, macrophages, monocytes, cycling myeloid cells and dendric cells) are highly enriched in immune-related domains 13 and 4, while the cancer-associated fibroblast (CAF) subpopulations[61] are highly enriched in the stroma domain 1.

Finally, we carefully examined the status of two hormonal receptors (*ESR1* and *PGR*) and one human epidermal growth factor receptor (*ERBB2*) in the tumor regions to characterize invasiveness and intratumor heterogeneity (Methods). *ERBB2* is highly expressed in the IDC domains 8 and 5, more so than that in the DCIS domains 10 and 16, supporting the advanced metastasis-related properties[62] of the IDC domains (Fig. 6g). *ESR1* is mainly expressed in the IDC and DCIS regions, while *PGR* is lowly expressed in all domains. Classification for these cancer-related domains (domains 5, 8, 10 and 16) based on the expression of the three receptors thus confirmed that most regions are either *ERBB2*$^+$ (HER2$^+$) or double-positive *ERBB2*$^+$/*ESR1*$^+$ (HER2$^+$/ER$^+$) (Fig. 6h and Supplementary Fig. 67). The proportion of *ERBB2*$^+$/*ESR1*$^+$ BC in DCIS regions (10 and 16) is much higher than that in the IDC regions (5 and 8), consistent with refs. 63–65. Quantifications of the proportion of spatial locations in each domain that display a high expression of the tumor invasiveness marker genes (*AGR3* and *CENPF*) further confirmed that domain 8 is more invasive than domains 5, 10 and 16 (Fig. 6i and Methods). Specifically, *AGR3* is associated with low-histological-grade breast tumors[66] and the proportion of spatial locations marked with high *AGR3* expression is higher in the DCIS domains 16 and 10. In contrast, *CENPF* is associated with tumor aggressive features and poor prognosis of patients with BC[67], and the proportion of spatial locations marked with high *CENPF* expression is higher in IDC domains 8 and 5. Similar results were observed in other tissue slices (Supplementary Figs. 68–71).

Discussion

We have presented a new computational method, IRIS, for accurate and scalable spatial domain detection in SRT studies via integrated reference-informed segmentation. IRIS is 4.6–666.0 times faster than existing methods in the moderate-sized datasets, while representing the only method scalable for large datasets including Stereo-seq and 10x Xenium (Supplementary Fig. 72 and Supplementary Table 12). We have demonstrated the benefits of IRIS through in-depth analysis of six SRT datasets generated from different technologies across distinct tissues and species. While, to our understanding, IRIS represents the first attempt to integrate a reference scRNA-seq with the SRT study to detect spatial domains, scRNA-seq has been previously paired with SRT for two other analytic settings[19,68,69], improving transcript spatial distribution prediction[70–74] and cell type proportion estimation[18,75,76]. The results here thus dovetail these recent findings and highlight the benefits of integrating reference scRNA-seq data to improve SRT analytics.

There are several important future extensions for IRIS. First, IRIS accommodates the cell type compositional similarity across

Fig. 5 | Analyzing the mouse olfactory bulb Stereo-seq subcellular data. **a**, The structure of the mouse olfactory bulb with the layers annotated on the basis of both the Allen Reference Atlas – Mouse Brain[78] and previous literature[79]. The mouse olfactory bulb is organized in an inside-out fashion and consists of SEL[46], GCL, IPL, MCL, EPL, GL, ONL and meninges[47]. The membrane surrounded the mouse olfactory bulb is called meninges. **b**, Spatial domains detected by IRIS in both slices. Details of the tissue slices that are used in this setting are presented in Supplementary Table 2. **c**, Bar plots displaying CHAOS values on the tissue slices of MOB when varying the prespecified number of spatial domains. The median CHAOSs across all slices ($n = 2$) in the analysis were used. **d**, Scatter plots displaying the spatial distribution of important MOB-related marker genes. **e**, Lollipop plots displaying the mean expression of important marker genes (y axis) in each spatial domain identified by IRIS (x axis). The solid arrow indicates the highest enrichment in the corresponding domain. **f**, The top enriched gene sets in selected spatial domains detected by IRIS (for example, domains 2 and 7) from the GSEA, with P values calculated on the basis of an adaptive multilevel split Monte-Carlo scheme implemented in the fgsea package[77]. **g**, A heat map plot displaying the estimated mean cell type proportion for representative cell types in each spatial domain detected by IRIS. The color scale was normalized to the 0–1 range. **h**, A spatial scatter plot displaying the spatial distribution of the IRIS-estimated cell type proportion for representative cell types across spatial locations. For **d**–**h**, the results are shown for the example S1 slice in the main analysis. OSN, olfactory sensory neuron; M-TC, mitral/tufted cell; GC, granule cell.

neighboring locations on each tissue slice through a graph Laplacian regularization by constructing an adjacency matrix, which can be easily formulated in a sparse form to further facilitate computation. IRIS can easily incorporate other types of similarity or kernel matrices such as the Gaussian kernels and the periodic kernels to capture the diverse and complex spatial correlation patterns that may be encountered

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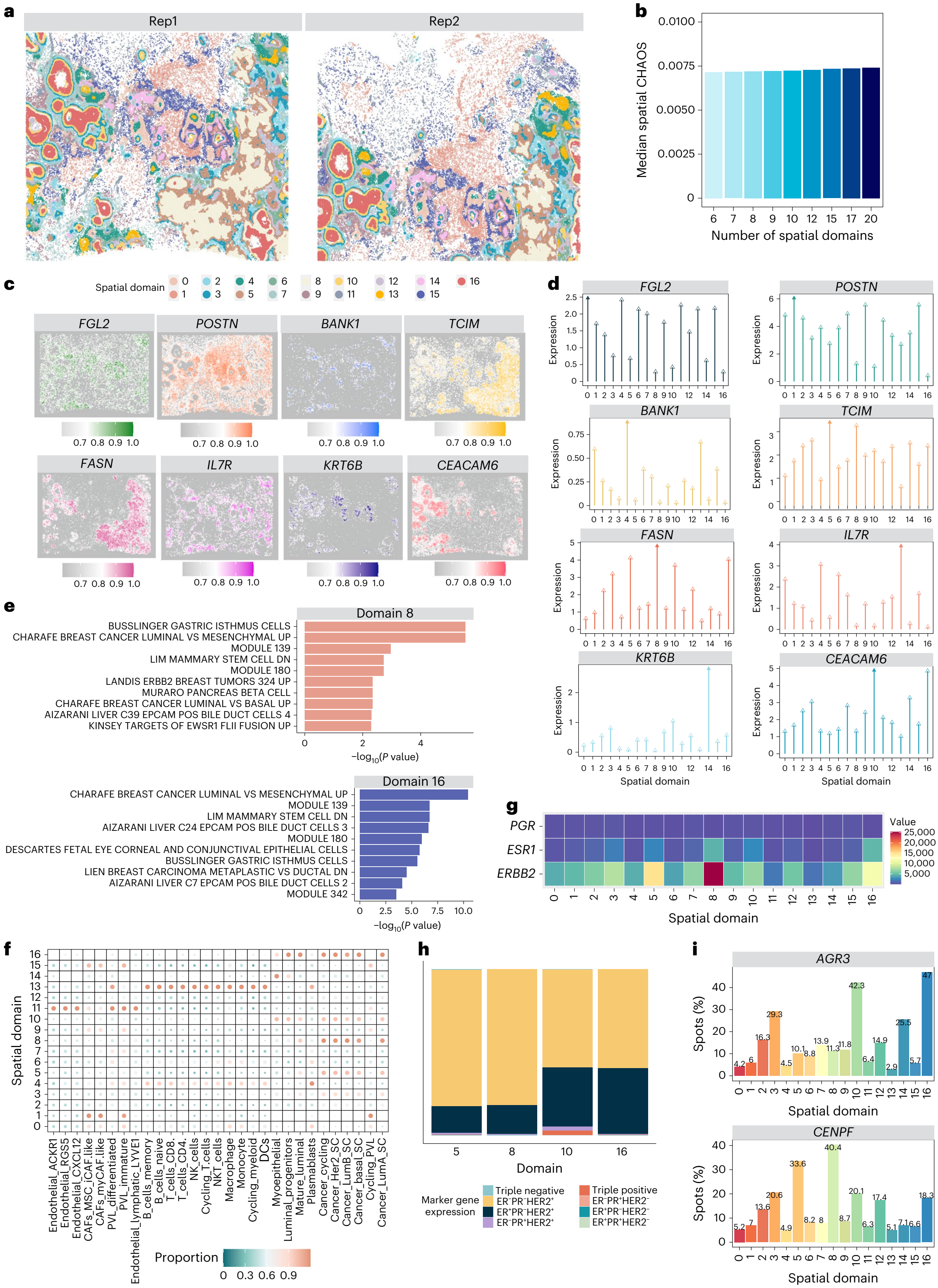

across datasets. IRIS can also incorporate a weighted combination of multiple kernel matrices to further improve performance. Second, we have primarily focused on using *k*-means clustering in IRIS because of its algorithmic stability and scalability. However, IRIS is general and can be paired with any other clustering algorithms such as Louvain or Leiden to take advantage of their distinct benefits. It is important to be

Fig. 6 | Analyzing the human BC 10x Xenium data. a, Spatial domains detected by IRIS in both slices. Details of the tissue slices used in this setting are provided in Supplementary Table 2. **b**, Bar plots displaying the CHAOS values on both tissue slices when varying the prespecified number of spatial domains. The median CHAOSs across all slices in the main analysis were used. **c**, Scatter plots displaying the spatial distribution of important BC-related marker genes. **d**, Lollipop plots displaying the mean expression of important marker genes (y axis) in each spatial domain identified by IRIS (x axis). The solid arrow indicates the highest enrichment value. **e**, The top enriched gene sets in the selected spatial domains identified by IRIS (for example, IDC domain 8 and DCIS domain 16) in the GSEA, with P values calculated on the basis of an adaptive multilevel split Monte-Carlo scheme implemented in the fgsea package[77]. **f**, A heat map plot displaying the estimated mean cell type proportion for representative cell types in each spatial domain detected by IRIS. The color scale was normalized to the 0–1 range. **g**, A heat map plot displaying the number of spatial locations that express two hormone receptors and one human epidermal growth factor receptor genes (*ERBB2*, *ESR2* and *PGR*) in each spatial domain detected by IRIS. **h**, A bar plot displaying the proportion of BC subtypes on the basis of these three receptors status in cancer domains detected by IRIS. **i**, A bar plot displaying the percentage of spatial locations that have high expression (greater than median expression across all spatial locations) of two important tumor invasiveness marker gene *AGR3* and *CENPF* in each spatial domain. For **c**–**i**, the results are shown for the example Rep1 slice in the main analysis. PVL, perivascular-like cell; NKT, natural killer T-like cell; DC, dendritic cells; LumA, Luminal A; LumB, Luminal B. Rep1 and Rep2 represent tissue replicates in the form of two sequential tissue slices.

aware, however, that using some of these algorithms may negatively impact computational performance. Third, we have mainly focused on using the transcriptomics measurements from both scRNA-seq and SRT studies. IRIS's modeling framework can in principle be extended to integrate other data modalities such as histological images that are often collected alongside SRT. However, such an extension of IRIS requires both image segmentation to identify cells on the tissue and cell alignment to align cells segmented on the image with SRT data, both of which remain technically challenging. Fourth, IRIS in principle could be extended to perform other SRT analytic tasks, such as mapping single cells from scRNA-seq onto spatial locations in SRT. Last, it is important to emphasize that IRIS is designed for spatial domain detection, not cell type deconvolution. We caution the over-interpretation of the cell type composition-informed low-dimensional components directly as the cell type composition (Supplementary Note 13).

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41592-024-02284-9.

References

1. Chen, K. H., Boettiger, A. N., Moffitt, J. R., Wang, S. & Zhuang, X. Spatially resolved, highly multiplexed RNA profiling in single cells. *Science* **348**, aaa6090 (2015).
2. Janesick, A. et al. High resolution mapping of the tumor microenvironment using integrated single-cell, spatial and in situ analysis. *Nat. Commun.* **14**, 8353 (2023).
3. Wang, X. et al. Three-dimensional intact-tissue sequencing of single-cell transcriptional states. *Science* **361**, eaat5691 (2018).
4. Ståhl, P. L. et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science* **353**, 78–82 (2016).
5. 10x Genomics: Visium Spatial Gene Expression. *10x Genomics* https://www.10xgenomics.com/ (2024).
6. Rodriques, S. G. et al. Slide-seq: a scalable technology for measuring genome-wide expression at high spatial resolution. *Science* **363**, 1463–1467 (2019).
7. Stickels, R. R. et al. Highly sensitive spatial transcriptomics at near-cellular resolution with Slide-seqV2. *Nat. Biotechnol.* **39**, 313–319 (2021).
8. Chen, A. et al. Spatiotemporal transcriptomic atlas of mouse organogenesis using DNA nanoball-patterned arrays. *Cell* **185**, 1777–1792. e1721 (2022).
9. Moses, L. & Pachter, L. Museum of spatial transcriptomics. *Nat. Methods* https://doi.org/10.1038/s41592-022-01409-2 (2022).
10. Rao, A., Barkley, D., França, G. S. & Yanai, I. Exploring tissue architecture using spatial transcriptomics. *Nature* **596**, 211–220 (2021).
11. Hu, J. et al. SpaGCN: integrating gene expression, spatial location and histology to identify spatial domains and spatially variable genes by graph convolutional network. *Nat. Methods* **18**, 1342–1351 (2021).
12. Zhao, E. et al. Spatial transcriptomics at subspot resolution with BayesSpace. *Nat. Biotechnol.* **39**, 1375–1384 (2021).
13. Li, Z. & Zhou, X. BASS: multi-scale and multi-sample analysis enables accurate cell type clustering and spatial domain detection in spatial transcriptomic studies. *Genome Biol.* **23**, 1–35 (2022).
14. Xu, H. et al. Unsupervised spatially embedded deep representation of spatial transcriptomics. *Genome Med.* **16**, 12 (2024).
15. Long, Y. et al. Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with GraphST. *Nat. Commun.* **14**, 1155 (2023).
16. Dong, K. & Zhang, S. Deciphering spatial domains from spatially resolved transcriptomics with an adaptive graph attention auto-encoder. *Nat. Commun.* **13**, 1739 (2022).
17. Allen, C., Chang, Y., Ma, Q. & Chung, D. MAPLE: a hybrid framework for multi-sample spatial transcriptomics data. Preprint at *bioRxiv* https://doi.org/10.1101/2022.02.28.482296 (2022).
18. Ma, Y. & Zhou, X. Spatially informed cell-type deconvolution for spatial transcriptomics. *Nat. Biotechnol.* https://doi.org/10.1038/s41587-022-01273-7 (2022).
19. Li, H. et al. A comprehensive benchmarking with practical guidelines for cellular deconvolution of spatial transcriptomics. *Nat. Commun.* **14**, 1548 (2023).
20. Bove, A. et al. Local cellular neighborhood controls proliferation in cell competition. *Mol. Biol. Cell* **28**, 3215–3228 (2017).
21. Maynard, K. R. et al. Transcriptome-scale spatial gene expression in the human dorsolateral prefrontal cortex. *Nat. Neurosci.* **24**, 425–436 (2021).
22. Mathys, H. et al. Single-cell transcriptomic analysis of Alzheimer's disease. *Nature* **570**, 332–337 (2019).
23. Huuki-Myers, L. et al. Integrated single cell and unsupervised spatial transcriptomic analysis defines molecular anatomy of the human dorsolateral prefrontal cortex. Preprint at *bioRxiv*. https://doi.org/10.1101/2023.02.15.528722 (2023).
24. Tang, Q. et al. Anatomical organization and spatiotemporal firing patterns of layer 3 neurons in the rat medial entorhinal cortex. *J. Neurosci.* **35**, 12346–12354 (2015).
25. Chung, W.-S. et al. Novel allele-dependent role for APOE in controlling the rate of synapse pruning by astrocytes. *Proc. Natl Acad. Sci. USA* **113**, 10186–10191 (2016).
26. Ocklenburg, S. et al. Myelin water fraction imaging reveals hemispheric asymmetries in human white matter that are associated with genetic variation in PLP1. *Mol. Neurobiol.* **56**, 3999–4012 (2019).
27. Swanson, O. K. & Maffei, A. From hiring to firing: activation of inhibitory neurons and their recruitment in behavior. *Front. Mol. Neurosci.* **12**, 168 (2019).

28. Ji, A. L. et al. Multimodal analysis of composition and spatial architecture in human squamous cell carcinoma. *Cell* **182**, 497–514. e422 (2020).
29. Yano, S., Tazawa, H., Kagawa, S., Fujiwara, T. & Hoffman, R. M. FUCCI real-time cell-cycle imaging as a guide for designing improved cancer therapy: a review of innovative strategies to target quiescent chemo-resistant cancer cells. *Cancers* **12**, 2655 (2020).
30. Pastushenko, I. & Blanpain, C. EMT transition states during tumor progression and metastasis. *Trends Cell Biol.* **29**, 212–226 (2019).
31. Kim, J. H. & Kim, N. Regulation of NFATc1 in osteoclast differentiation. *J. bone Metab.* **21**, 233–241 (2014).
32. Olson, E., Geng, J. & Raghavan, M. Polymorphisms of HLA-B: influences on assembly and immunity. *Curr. Opin. Immunol.* **64**, 137–145 (2020).
33. Chen, H. et al. Dissecting mammalian spermatogenesis using spatial transcriptomics. *Cell Rep.* **37**, 109915 (2021).
34. Green, C. D. et al. A comprehensive roadmap of murine spermatogenesis defined by single-cell RNA-seq. *Dev. Cell* **46**, 651–667. e610 (2018).
35. Linn, E., Ghanem, L., Bhakta, H., Greer, C. & Avella, M. Genes regulating spermatogenesis and sperm function associated with rare disorders. *Front. Cell Dev. Biol.* **9**, 634536 (2021).
36. Ren, S. et al. The expression, function, and utilization of Protamine1: a literature review. *Transl. Cancer Res.* **10**, 4947 (2021).
37. Yan, W. et al. Zmynd15 encodes a histone deacetylase-dependent transcriptional repressor essential for spermiogenesis and male fertility. *J. Biol. Chem.* **285**, 31418–31426 (2010).
38. Lehtiniemi, T. & Kotaja, N. in *Genetics of Human Infertility* Vol. 21 (eds Vogt, P. H. & Schmid, K.) 101–115 (2017).
39. Wang, X. et al. BET bromodomain inhibitor JQ1 regulates spermatid development by changing chromatin conformation in mouse spermatogenesis. *Genes Dis.* **9**, 1062–1073 (2022).
40. Nguyen, A. L. & Schindler, K. Specialize and divide (twice): functions of three aurora kinase homologs in mammalian oocyte meiotic maturation. *Trends Genet.* **33**, 349–363 (2017).
41. Zhang, L. et al. Reprogramming of Sertoli cells to fetal-like Leydig cells by Wt1 ablation. *Proc. Natl Acad. Sci. USA* **112**, 4003–4008 (2015).
42. Ding, G.-L. et al. The effects of diabetes on male fertility and epigenetic regulation during spermatogenesis. *Asian J. Androl.* **17**, 948 (2015).
43. Antonetti, D. A., Reynet, C. & Kahn, C. R. Increased expression of mitochondrial-encoded genes in skeletal muscle of humans with diabetes mellitus. *J. Clin. Investig.* **95**, 1383–1388 (1995).
44. Al-Kafaji, G., Sabry, M. A. & Bakhiet, M. Increased expression of mitochondrial DNA-encoded genes in human renal mesangial cells in response to high glucose-induced reactive oxygen species. *Mol. Med. Rep.* **13**, 1774–1780 (2016).
45. Tepe, B. et al. Single-cell RNA-seq of mouse olfactory bulb reveals cellular heterogeneity and activity-dependent molecular census of adult-born neurons. *Cell Rep.* **25**, 2689–2703. e2683 (2018).
46. Nagayama, S., Homma, R. & Imamura, F. Neuronal organization of olfactory bulb circuits. *Front. Neural Circuits* **8**, 98 (2014).
47. Gudjohnsen, S. A. et al. Meningeal melanocytes in the mouse: distribution and dependence on Mitf. *Front. Neuroanat.* **9**, 149 (2015).
48. DeSisto, J. et al. Single-cell transcriptomic analyses of the developing meninges reveal meningeal fibroblast diversity and function. *Dev. Cell* **54**, 43–59. e44 (2020).
49. Chen, Y. et al. A single-cell atlas of mouse olfactory bulb chromatin accessibility. *J. Genet. Genomics* **48**, 147–162 (2021).
50. Li, C. et al. Identifying potential diagnostic genes for diabetic nephropathy based on hypoxia and immune status. *J. Inflamm. Res.* **14**, 6871 (2021).
51. Zeppilli, S. et al. Molecular characterization of projection neuron subtypes in the mouse olfactory bulb. *eLife* **10**, e65445 (2021).
52. Wang, D. et al. VIP interneurons regulate olfactory bulb output and contribute to odor detection and discrimination. *Cell Rep.* **38**, 110383 (2022).
53. Hawrylycz, M. et al. in *Springer Handbook of Bio-/Neuroinformatics* 1111–1126 (Springer, 2014).
54. Haslinger, A., Schwarz, T. J., Covic, M. & Chichung Lie, D. Expression of Sox11 in adult neurogenic niches suggests a stage-specific role in adult neurogenesis. *Eur. J. Neurosci.* **29**, 2103–2114 (2009).
55. Young, J. K., Heinbockel, T. & Gondré-Lewis, M. C. Astrocyte fatty acid binding protein-7 is a marker for neurogenic niches in the rat hippocampus. *Hippocampus* **23**, 1476–1483 (2013).
56. Fujiwara, N. & Cave, J. W. Partial conservation between mice and humans in olfactory bulb interneuron transcription factor codes. *Front. Neurosci.* **10**, 337 (2016).
57. Sun, X., Liu, X., Starr, E. R. & Liu, S. CCKergic tufted cells differentially drive two anatomically segregated inhibitory circuits in the mouse olfactory bulb. *J. Neurosci.* **40**, 6189–6206 (2020).
58. Wu, S. Z. et al. A single-cell and spatially resolved atlas of human breast cancers. *Nat. Genet.* **53**, 1334–1347 (2021).
59. Su, K. et al. TC-1 (c8orf4) enhances aggressive biologic behavior in lung cancer through the Wnt/β-catenin pathway. *J. Surg. Res.* **185**, 255–263 (2013).
60. Han, S. et al. CEACAM5 and CEACAM6 are major target genes for Smad3-mediated TGF-β signaling. *Oncogene* **27**, 675–683 (2008).
61. Von Ahrens, D., Bhagat, T. D., Nagrath, D., Maitra, A. & Verma, A. The role of stromal cancer-associated fibroblasts in pancreatic cancer. *J. Hematol. Oncol.* **10**, 1–8 (2017).
62. Yu, D. & Hung, M.-C. Overexpression of ErbB2 in cancer and ErbB2-targeting strategies. *Oncogene* **19**, 6115–6121 (2000).
63. Miligy, I. M. et al. The clinical and biological significance of HER2 over-expression in breast ductal carcinoma in situ: a large study from a single institution. *Br. J. Cancer* **120**, 1075–1082 (2019).
64. Thorat, M. A. et al. Prognostic and predictive value of HER2 expression in ductal carcinoma in situ: results from the UK/ANZ DCIS randomized trial. *Clin. Cancer Res.* **27**, 5317–5324 (2021).
65. Bergholtz, H. et al. Contrasting DCIS and invasive breast cancer by subtype suggests basal-like DCIS as distinct lesions. *NPJ Breast Cancer* **6**, 26 (2020).
66. Jian, L. et al. AGR3 promotes estrogen receptor-positive breast cancer cell proliferation in an estrogen-dependent manner. *Oncol. Lett.* **20**, 1441–1451 (2020).
67. O'Brien, S. L. et al. CENP-F expression is associated with poor prognosis and chromosomal instability in patients with primary breast cancer. *Int. J. Cancer* **120**, 1434–1443 (2007).
68. Li, B. et al. Benchmarking spatial and single-cell transcriptomics integration methods for transcript distribution prediction and cell type deconvolution. *Nat. Methods* **19**, 662–670 (2022).
69. Sun, D., Liu, Z., Li, T., Wu, Q. & Wang, C. STRIDE: accurately decomposing and integrating spatial transcriptomics using single-cell RNA sequencing. *Nucleic Acids Res.* **50**, e42 (2022).
70. Nitzan, M., Karaiskos, N., Friedman, N. & Rajewsky, N. Gene expression cartography. *Nature* **576**, 132–137 (2019).
71. Cang, Z. & Nie, Q. Inferring spatial and signaling relationships between cells from single cell transcriptomic data. *Nat. Commun.* **11**, 1–13 (2020).
72. Abdelaal, T., Mourragui, S., Mahfouz, A. & Reinders, M. J. SpaGE: spatial gene enhancement using scRNA-seq. *Nucleic Acids Res.* **48**, e107–e107 (2020).

https://doi.org/10.1038/s41592-024-02284-9

73. Lopez, R. et al. A joint model of unpaired data from scRNA-seq and spatial transcriptomics for imputing missing gene expression measurements. Preprint at *arXiv* https://doi.org/10.48550/arXiv.1905.02269 (2019).
74. Biancalani, T. et al. Deep learning and alignment of spatially resolved single-cell transcriptomes with Tangram. *Nat. Methods* **18**, 1352–1362 (2021).
75. Cable, D. M. et al. Robust decomposition of cell type mixtures in spatial transcriptomics. *Nat. Biotechnol.* **40**, 517–526 (2022).
76. Kleshchevnikov, V. et al. Cell2location maps fine-grained cell types in spatial transcriptomics. *Nat. Biotechnol.* **40**, 661–671 (2022).
77. Korotkevich, G. et al. Fast gene set enrichment analysis. Preprint at *bioRxiv* https://doi.org/10.1101/060012 (2021).
78. Allen Reference Atlas – Mouse Brain [brain atlas]. *Allen Institute for Brain Science* https://atlas.brain-map.org/ (2024).
79. Imamura, F., Ito, A. & LaFever, B. J. Subpopulations of projection neurons in the olfactory bulb. *Front. Neural Circuits* **14**, 561822 (2020).

Methods

IRIS method overview

We present an overview of IRIS here, with its technical details provided in Supplementary Note 1. IRIS requires two types of data input: scRNA-seq data that serve as the reference and SRT data that measure the transcriptomic profiles on multiple tissue slices. The scRNA-seq reference data consist of K cell types with a set of G cell-type-informative genes, which can be selected on the basis of ref. 18. We follow refs. 18,80 to extract from the scRNA-seq data a reference basis matrix B, which is a G-by-K matrix that contains the mean expression level of the G cell-type-informative genes in the K annotated cell types. The SRT data consist of T different tissue slices ($T \geq 2$), and we denote Y_t as the G-by-N_t gene expression matrix in slice t for the same set of G informative genes measured on N_t spatial locations. We assume that there is a total of R distinct spatial domains across T tissue slices. We introduce an N_t vector of spatial domain indicators $\mathbf{c}_t$ to indicate the domain label for each spatial location on tissue slice t, where each element of $\mathbf{c}_t$ takes values in $\{1, \ldots, R\}$.

Our goal is to detect the spatial domains on the tissue slices in SRT by inferring the spatial domain indicators $\mathbf{c}_t$. Inferring $\mathbf{c}_t$ requires knowing the cell type composition of each spatial location on the tissue slice, since every spatial domain is characterized by a unique composition of cell types. Therefore, to facilitate the inference on $\mathbf{c}_t$, we introduce an N_t-by-K cell type composition-informed low-dimensional components matrix P_t, where each row of P_t represents either the compositions of the K cell type (for low-resolution spot-level SRT) or the probabilities of the K cell type assignments (for single-cell or subcellular-resolution SRT) to each spatial location on slice t (detailed interpretation on P_t for high-resolution SRT data in Supplementary Note 13).

The composition matrix P_t is not only key for inferring $\mathbf{c}_t$ but also serves as an important link between the reference basis matrix B in scRNA-seq and the expression matrix Y_t in SRT, thus connecting the two distinct data types. In particular, each element of Y_t, which describes the expression level of an informative gene on a spatial location, can be expressed as the product of the gene's expression level in each cell type and the cell type compositions on the location. Consequently, we can infer P_t by minimizing the difference between Y_t and BP_t^T in terms of Euclidean distance (also known as square of the Frobenius norm) through the cost function

$$\min_{\substack{0 \leq P_t \leq 1 \\ t = 1, 2, \ldots T}} \sum_{t=1}^{T} \left\| Y_t - BP_t^T \right\|_F^2. \tag{1}$$

In the process, we constrain each element of P_t to be nonnegative and we accommodate the spatial correlation in the cell type composition among neighboring locations, which are commonly observed on each tissue slice, through the penalty function

$$\sum_{t=1}^{T} \mathrm{Tr}\left(P_t^T L_t P_t\right), \tag{2}$$

where $\mathrm{Tr}(\cdot)$ denotes the trace of a matrix and L_t is the graph Laplacian matrix for slice t, expressed as the difference between two matrices $L_t = D_t - A_t$. Here, A_t is an N_t-by-N_t adjacency matrix[81,82]: its ijth element is 1 if the ith and jth locations on slice t are mutual neighbors, defined as being the k nearest neighbors of each other (default $k = 10$), and is zero otherwise. D_t is a diagonal matrix whose entries are column sums of A_t. Minimizing each term $\mathrm{Tr}(P_t^T L_t P_t)$ is equivalent to minimizing the summation of the weighted square of the Euclidean norm $\frac{1}{2}\sum_{i,j=1}^{N_t} \left\| P_{t_i}^T - P_{t_j}^T \right\|_2^2 A_{t_{ij}}$, where P_{t_i} is the ith row of P_t and $A_{t_{ij}}$ is the ijth element of A_t (details in Supplementary Note 1). Therefore, minimizing the penalty function in equation (2) encourages similarity in the cell type composition in the neighboring locations on each slice, facilitating accurate and robust inference of P_t. Through modeling spatial correlations, the penalty function of equation (2) also allows us to effectively integrate the SRT profiles across neighboring spatial locations on each tissue slice.

Besides connecting the SRT data with the reference scRNA-seq data, the composition matrix P_t also provides direct evidence for inferring the spatial domain indicators $\mathbf{c}_t$. Specifically, we consider the following k-means-type cost function that connects P_t to $\mathbf{c}_t$:

$$\sum_{t=1}^{T} \sum_{r=1}^{R} \left\| \frac{1}{n_{t_r}} P_t^T \mathbf{q}_{t_r} - \mu_r \right\|_2^2, \tag{3}$$

where $\mathbf{q}_{t_r}$ is an N_t vector of indicator variables, $\mathbf{q}_{t_r}(i) = \begin{cases} 1, & \text{if } \mathbf{c}_{t_i} = r \\ 0, & \text{otherwise} \end{cases}$, with each ith element being 1 when the corresponding location i belongs to the rth spatial domain on tissue slice t and being zero otherwise; n_{t_r} is the total number of locations in the spatial domain r on tissue slice t (that is, $n_{t_r} = \sum_i \mathbf{q}_{t_r}(i)$); and $\boldsymbol{\mu}_r$ is a K vector of domain-specific cell type composition-informed low-dimensional components profile for the rth spatial domain. Equation (3) directly relates the average cell type compositions for locations residing on the same spatial domain across tissue slices to the common cell type composition profile parameter $\boldsymbol{\mu}_r$, thus encouraging similarity in the domain-specific cell type compositions across tissue slices. Encouraging such similarity allows us to borrow information across multiple slices for accurate and robust spatial domain inference.

The above equations (1)–(3) characterize different aspects of the IRIS modeling framework. In particular, equation (1) enables the integration of the transcriptomic profiles between scRNA-seq and SRT data. Equation (2) encourages within-slice cell type compositional similarities among neighboring locations and allows for integrative transcriptomics analysis across locations on each tissue slice. Equation (3) models the consistency of cell type composition on the same spatial domain across slices, allowing for integrative transcriptomics analysis across multiple tissue slices. Importantly, we incorporate all three equations together into a joint cost function

$$\phi(P_t) = \sum_{t=1}^{T} \left\| Y_t - BP_t^T \right\|_2^2 + \beta \sum_{t=1}^{T} \sum_{r=1}^{R} \left\| \frac{1}{n_{t_r}} P_t^T \mathbf{q}_{t_r} - \mu_r \right\|_2 + \lambda \sum_{t=1}^{T} \mathrm{Tr}\left(P_t^T L_t P_t\right), \tag{4}$$

where β and λ determine the relative contribution of the three components. The scale of the two parameters β and λ is determined by the dimensionality of the matrices in the three components, which are in turn determined by N_t, G and K. We set β and λ to fixed values of 1,000 and 2,000 throughout the present study, with robustness analysis provided in Supplementary Fig. 73. Note that equation (1) involves summation of G by N_t terms and is thus several orders of magnitude larger than equation (2), which involves summation of K terms, or equation (3), which involves summation of K-by-K terms. Consequently, to achieve effective regularization, the hyperparameters β and λ have to be set to high values so that equations (2) and (3) are not completely incomparable with equation (1) (Supplementary Note 14).

In the joint model defined in equation (4), our primary parameters of interest are the spatial domain labels for the locations on every tissue slice, $\mathbf{c}_t$, for $t \in \{1, \ldots, T\}$. For model inference, we develop a constrained iterative optimization algorithm, which iteratively updates the spatial domain labels $\mathbf{c}_t$, along with the secondary parameters that include the composition matrix P_t and the domain-specific composition profile $\boldsymbol{\mu}_r$ (details in Supplementary Note 1). Briefly, IRIS performs optimization on the joint cell type composition matrices and the spatial domain label matrices in an iterative fashion, with nonnegativity constraints on each element of the composition matrix P_t. In the process, we utilize a k-means unsupervised clustering algorithm on the concatenated composition matrix P_t to update the spatial domain label $\mathbf{c}_t$. Given the updated $\mathbf{c}_t$, IRIS then updates the secondary parameters that include the composition matrix P_t and the domain-specific composition profile $\mathbf{\mu}_r$. This interdependent iterative refinement of all parameters ensures that the identified spatial domains are characterized by the uniquely defined composition profiles, thereby enhancing the accuracy of spatial domain detection (Supplementary Note 1).

Compared methods for spatial domain detection

In the benchmarking human DLPFC dataset, we compared the performance of IRIS with eight methods belonging to three categories: (1) methods that analyze a single slice at a time (single-slice-based methods): spaGCN (version 1.2.5, with or without image), BayesSpace (version 1.5.1), SEDR (version 1.0.0, downloaded on 4 January 2022), GraphST (version 1.1.1) and STAGATE (version 1.0.1); (2) methods that jointly analyze multiple slices (multislice-based methods): BayesSpaceJoint, which is a variation of BayesSpace (version 1.5.1), BASS (version 1.1.0.16), Maple (version 0.99.1), GraphSTJoint (version 1.1.1) and STAGATEJoint (version 1.0.1); (3) methods that first infer cell type compositions on the tissue and then conduct tissue segmentation via k-means clustering on the inferred compositions (deconvolution-based methods): CARD (version 1.0) and RCTD (version 1.1.0). Specifically, we mainly follow the corresponding GitHub tutorials to run the above methods for spatial domain detection, with details in Supplementary Note 15.

Real data analyses

We applied IRIS to analyze six published SRT datasets collected by different techniques, with distinct spatial resolutions, and from multiple species and tissues. For all SRT data, we obtained an external scRNA-seq collected on the same type of tissue but with a different sequencing technology to serve as the reference (Supplementary Table 1). Details of preprocessing are in Supplementary Note 16.

Spatial domain detection. We compared IRIS's performance with other spatial domain detection methods, using the same SRT data as input for all methods (Supplementary Tables 1 and 2).

In the DLPFC dataset by 10x Visium, we examined five settings in total: (1) baseline setting when the slices are from the same donor that share high similarity; (2) challenging setting when we use a scRNA-seq reference with one missing cell type information at a time; (3) challenging setting when we use a scRNA-seq reference with mis-classified cell type information by randomly merging two cell types; (4) challenging setting when we used an alternative scRNA-seq reference data from a separate study[23] with distinct cell type number, annotation and resolution (Supplementary Fig. 5); (5) challenging setting when the slices are from different donors that share low similarity in structures (Supplementary Note 4). In each setting, we used the annotated tissue domains as the ground truth and evaluated method performance by calculating the ARI (Supplementary Note 2). In addition to use the ground truth, we also evaluated method performance using marker gene enrichment (details in the next section). For the marker gene enrichment quantification and other downstream analysis, we focus on the slice 151509 as the main example and provide the analysis results for other tissue slices in Supplementary Figs. 14–16.

In the SCC dataset by ST, we followed the original publication[28] to set the number of spatial domains to be 12 for the tissue slices from patients 2, 5 and 10. We further varied the number of spatial domains from 8 to 14 to evaluate their performance. This dataset contains a hematoxylin and eosin (H&E) image with annotated leading edges for one tissue slice (P2_ST_rep2). For this slice, we extracted the annotations for the tumor and nontumor regions based on the leading edges and treated the extracted regional annotations as coarse ground. We then fitted a logistic regression model, with inferred spatial domain labels as predictors to predict the extracted tumor versus nontumor annotations, and evaluate the method accuracy through pseudo-R^2 (Nagelkerke and MacFadden). Besides these annotations, method performance was also assessed using CHAOS and marker gene enrichment quantification within tumor core and edge regions. We focus on the slice P2_ST_rep2 as the main example and provide the analysis results for the other tissue slices in the Supplementary Figs. 24 and 25.

In the high-resolution mouse spermatogenesis data by Slide-seq, we set the number of spatial domains to be 5 and varied it from 4 to 10 to evaluate method performance. Because no histology images or manual annotations were available, we relied on the well-defined seminiferous tubule structures in mouse testes to evaluate method performance. In addition to carefully examining the inferred tubule structures, we also evaluated method performance using the CHAOS and marker gene enrichment. We focus on the slice WT3_Puck7 as the main example and perform downstream analysis to compare the testis tissue structure of WT mouse tissue slices versus that of diabetic mouse tissue slices.

In the single-cell-resolution mouse brain data by Vizgen MERFISH, we set the number of domain clusters to be 31 and varied it from 10 to 31 to evaluate method performance. Because no histology image nor manual domain annotations were available, we evaluated method performance by calculating CHAOS and marker gene enrichment quantification in the known hippocampus regions. We focus on the slice S2R3 as the main example and provide the analysis results for other tissue slices in the Supplementary Figs. 49–52.

For the subcellular mouse olfactory bulb (MOB) data by Stereo-seq, we set the number of spatial domains to be 11 based on the domain knowledge of the MOB structure. We varied the number of spatial domains from 4 to 14, and we evaluated the performance of IRIS by calculating CHAOS and marker gene enrichment. We focus on slice S1 as the main example and provide the analysis results for other tissue slice in the Supplementary Fig. 60. Note that none of the other methods was scalable to these large data.

For the high-resolution human BC data by 10x Xenium, we followed the original publication to set the number of spatial domains to be 17 (ref. 2). We varied the number of spatial domains from 6 to 20, and we evaluated the performance of IRIS by calculating CHAOS and marker gene enrichment. We focus on the slice Rep1 as the main example and provide the analysis results for the other tissue slice in the Supplementary Figs. 68–71. Note that none of the other methods was scalable to these large data.

Domain-specific marker gene quantification across different methods. We quantified the performance of different methods by comparing the fold change of marker genes in their corresponding spatial domains in the first three datasets (the two large datasets were excluded as the other methods cannot deal with them). We reasoned that a good spatial domain detection method would yield accurate tissue structures that capture the enrichment of domain-specific marker genes. Because the spatial domain labels inferred from different methods are arbitrary (that is, domain 1 from one method does not necessarily corresponds to domain 1 from another method), to ensure fair comparison, we mapped the domain labels of each method onto a common label system so that the same domain label from different methods corresponds to the same anatomic region (that is, domain 1 from all methods now corresponds to the same anatomic region; details in Supplementary Note 17). Afterwards, with a corresponding set of spatial domain labels, we calculate the fold change of each marker gene in their expected spatial domain to evaluate the performance of different methods. For the human DLPFC dataset, we excluded the deconvolution-based methods (that is, CARD_kmeans and RCTD_kmeans) and Maple, as these three methods cannot capture the layered structure of DLPFC tissue. For the human SCC dataset, we compared only the spaGCN, BayesSpace and BayesSpaceJoint, as the other methods fail to detect the tumor core and edge regions. For the Slide-seq mouse testis dataset, we compared only the BayesSpaceJoint, SEDR, BASS and IRIS for marker gene enrichment, as the other methods could not capture the tubular structures in all the WT and ob/ob tissue slices. For the mouse brain Vizgen MERFISH data, we did not perform such a comparison as it is difficult to correctly map 31 spatial regions into a common coordinate system. Instead, we carefully compare the spatial domains detected by different methods with known mouse brain anatomic structures provided by Allen's Institute.

DE and GSEA. For the human SCC data, we focused on the tumor core and edge regions. We performed DE analysis and GSEA between domains 6 and 3, and between domains 8 and 6 to reveal the tumor regional heterogeneity. Specifically, for each set of DE analysis, we first conducted a Wilcoxon rank sum test by using Seurat[83] with the function wilcoxauc[83,84]. We declared a gene to be an upregulated DE gene if its Benjamini–Hochberg (BH)-adjusted *P* value <0.05 and its log fold change >0. We declared a gene to be a downregulated DE gene if its BH-adjusted *P* value <0.05 and log fold change <0. GSEA was performed by fgsea[77], a method for fast preranked GSEA. For the other datasets, we performed DE analysis and GSEA analysis in a domain-specific fashion by using Seurat[83] with the function wilcoxauc[83,84]. We declared a gene to be a significant domain-specific gene if its BH-adjusted *P* value <0.01 and log fold change >0. We extracted corresponding gene sets from the R package msigdbr[85]. For the human datasets (that is, human SCC, DLPFC and BC data), we focused on testing all human gene sets. For the mouse datasets (that is, mouse spermatogenesis, brain and MOB data), we focused on testing mouse ontology gene sets (C5). We declared gene sets to be significantly enriched on the basis of a BH-adjusted p-value threshold of 0.05.

Additional analyses include the nearest-neighbor analysis in the human SCC ST data, and purity score analysis in the mouse spermatogenesis Slide-seq data is listed in Supplementary Note 17.

Human BC subtype analysis

We classified the human BC 10x Xenium spatial locations on the basis of the expression of two important hormone receptors *ESR1* (estrogen receptor) and *PGR* (progesterone receptor) and the epidermal growth factor receptor *ERBB2* (also known as *HER2*). These three receptors have been used to classify cancer subtypes and to compare the DCIS region with the IDC region[86–89]. Additionally, $ERBB2^+/ESR1^+$ cancer subtypes are less invasive[90,91] and ESR1 expression influences the immune cell composition and the immune landscape of early-stage breast tumors[92]. For location classification, we divided each location into eight subtypes based on whether each of the three genes were expressed or not: triple negative, $ER^+/PR^-/HER2^-$, $ER^-/PR^+/HER2^-$, $ER^-/PR^-/HER2^+$, $ER^+/PR^+/HER2^-$, $ER^+/PR^-/HER2^+$, $ER^-/PR^+/HER2^+$, triple positive. We then calculated the percentage of the eight subtypes in the tumor domains detected by IRIS (domains 5, 8, 10 and 16). In addition, we used two important tumor invasiveness marker genes *AGR3* and *CENPF* to quantify the invasiveness of the aforementioned four tumor domains. Specifically, for each domain, we calculated the percentage of its spatial locations whose marker gene expression is higher than the median level across all locations. Such a percentage serves as the evidence for whether the expression of *AGR3* or *CENF* is high in the spatial domain.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The original public data used in this work can be accessed through the following links: human DLPFC data by 10x Visium available at http://spatial.libd.org/spatialLIBD/, with human post-mortem brain single-nucleus RNA-seq reference data available at Synapse (https://www.synapse.org/#!Synapse:syn18485175); human SCC data by ST are available at GEO accession GSE144240, with the human SCC scRNA-seq reference data available at GEO accession GSE144236; mouse spermatogenesis data by Slide-seq are available at https://www.dropbox.com/s/ygzpj0d0oh67br0/Testis_Slideseq_Data.zip?dl=0, with the mouse testis scRNA-seq reference data available at GEO accession GSE112393; mouse brain (coronal section) Vizgen MERFISH data are available at https://info.vizgen.com/mouse-brain-data, with the mouse brain scRNA-seq reference data available at http://mousebrain.org/adolescent/; mouse olfactory bulb by Stereo-seq data are available at https://db.cngb.org/stomics/mosta/download/, with the mouse olfactory bulb scRNA-seq data available at GEO accession GSE121891; human BC by 10x Xenium data are available at https://www.10xgenomics.com/products/xenium-in-situ/preview-dataset-human-breast, with the human BC scRNA-seq reference data available at GSE accession GSE176078; details about the data we used in this study are provided in Supplementary Tables 1 and 2.

Code availability

The IRIS software package and source code have been deposited at https://xiangzhou.github.io/software/ and https://github.com/YingMa0107/IRIS. All scripts used to reproduce all the analysis are also available at the same website.

References

80. Wang, X., Park, J., Susztak, K., Zhang, N. R. & Li, M. Bulk tissue cell type deconvolution with multi-subject single-cell expression reference. *Nat. Commun.* **10**, 1–9 (2019).
81. Chung, F. R. & Graham, F. C. *Spectral Graph Theory* Vol. 92 (American Mathematical Society, 1997).
82. Grone, R., Merris, R. & Sunder, V. The Laplacian spectrum of a graph. *SIAM J. Matrix Anal. Appl.* **11**, 218–238 (1990).
83. Argelaguet, R., Cuomo, A. S., Stegle, O. & Marioni, J. C. Computational principles and challenges in single-cell data integration. *Nat. Biotechnol.* **39**, 1202–1215 (2021).
84. Korsunsky, I., Nathan, A., Millard, N. & Raychaudhuri, S. Presto scales Wilcoxon and auROC analyses to millions of observations. Preprint at *bioRxiv* https://doi.org/10.1101/653253 (2019).
85. Dolgalev, I. msigdbr: MSigDB gene sets for multiple organisms in a tidy data format. R package version 7. *R Project* https://cran.r-project.org/web/packages/msigdbr/msigdbr.pdf (2020).
86. Peres, R. M. R. et al. Comparative evaluation of the erbB2 and hormone receptor status of neighboring invasive and in situ components of ductal carcinomas of the breast. *Int. J. Biol. Markers* **24**, 238–244 (2009).
87. Villanueva, H. et al. The emerging roles of steroid hormone receptors in ductal carcinoma in situ (DCIS) of the breast. *J. Mammary Gland Biol. Neoplasia* **23**, 237–248 (2018).
88. Hussein, M. R., Abd-Elwahed, S. R. & Abdulwahed, A. R. Alterations of estrogen receptors, progesterone receptors and c-erbB2 oncogene protein expression in ductal carcinomas of the breast. *Cell Biol. Int.* **32**, 698–707 (2008).
89. Goh, C. W. et al. Invasive ductal carcinoma with coexisting ductal carcinoma in situ (IDC/DCIS) versus pure invasive ductal carcinoma (IDC): a comparison of clinicopathological characteristics, molecular subtypes, and clinical outcomes. *J. Cancer Res. Clin. Oncol.* **145**, 1877–1886 (2019).
90. Harada, S. et al. The significance of HER-2/neu receptor positivity and immunophenotype in ductal carcinoma in situ with early invasive disease. *J. Surg. Oncol.* **104**, 458–465 (2011).
91. Miligy, I. M. et al. The clinical significance of oestrogen receptor expression in breast ductal carcinoma in situ. *Br. J. Cancer* **123**, 1513–1520 (2020).
92. Voorwerk, L. et al. Immune landscape of breast tumors with low and intermediate estrogen receptor expression. *NPJ Breast Cancer* **9**, 39 (2023).

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Author contributions

Y.M. and X.Z. conceived the idea and designed the study. Y.M. developed the method, implemented the software and analyzed real data. Y.M. and X.Z. wrote the manuscript, and all authors read and approved the final manuscript.

Competing interests

The authors declare that they have no competing interests.

Additional information

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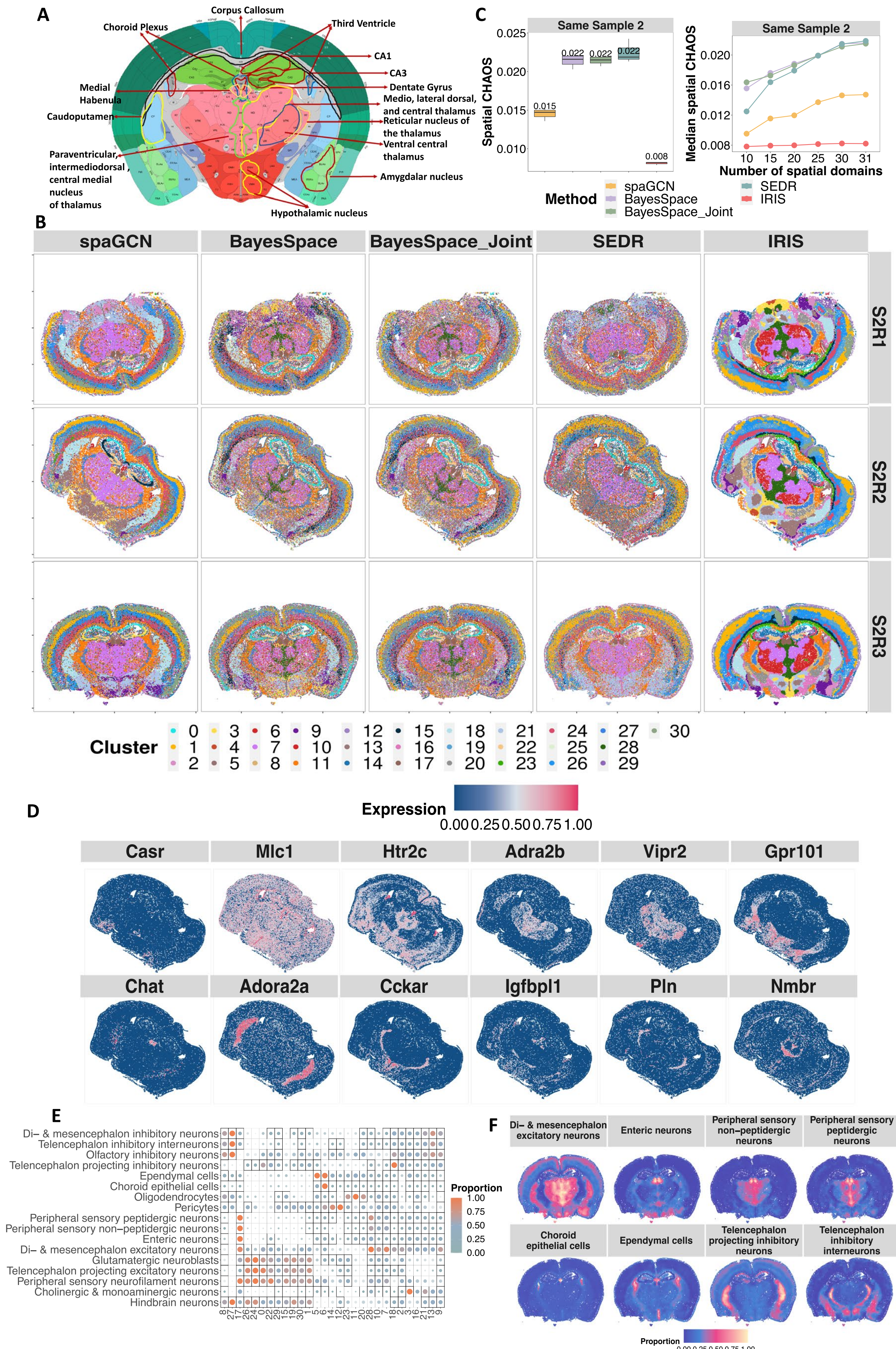

Extended Data Fig. 1 | See next page for caption.

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Extended Data Fig. 1 | Analyzing the mouse brain Vizgen MERFISH single-cell data. **(a)** The structure of the mouse brain with the main tissue regions annotated from the Allen Reference Atlas – Mouse Brain. **(b)** Spatial domains detected by IRIS, spaGCN, BayesSpace, BayesSpaceJoint, and SEDR in the 'Sample 2' analysis. Details of the tissue slices used in this setting are provided in Supplementary Table 2. **(c)** Boxplots display CHAOS values for different methods, which measure the spatial continuity and compactness of the detected spatial domains from different methods, on the tissue replicates of slice 2 (n = 2, left panel). Each boxplot ranges from the first and third quartiles with the median as the horizontal line while whiskers represent 1.5 times the interquartile range from the lower and upper bounds of the box. Compared spatial domain detection methods (x-axis) include spaGCN (yellow), BayesSpace (purple), BayesSpaceJoint (green), SEDR (blue), and IRIS (red). Line plots display CHAOS values when varying the pre-specified number of spatial domains. The median CHAOSs across all 3 tissue replicates of regional sample 2 was calculated. **(d)** Scatter plots display the spatial distribution of important brain tissue related marker genes. **(e)** Heatmap plot displays the estimated mean cell type proportion for representative cell types in each spatial domain detected by IRIS. Color scale was normalized to 0-1 range. **(f)** Spatial scatter plot displays the spatial distribution of IRIS estimated cell type proportion for representative cell types across spatial locations. For D – F, the results are shown for the example S2R3 slice in the main analysis.

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Software and code

Policy information about availability of computer code

Data collection	No software is used in data collecting
Data analysis	We used the newly developed R C++ package IRIS for data analysis. IRIS is described in the Methods section and deposited at Github [https://github.com/YingMa0107/IRIS]. In addition, we also used the following software for comparative analysis: (1) spaGCN (version 1.2.5, https://github.com/jianhuupenn/SpaGCN); (2) BayesSpace (version 1.5.1, https://github.com/edward130603/BayesSpace); (3) BayesSpaceJoint, which is a variation of BayesSpace (version 1.5.1) that first uses Harmony to remove batch effects and then jointly analyzes multiple tissue slices; and (4) SEDR (version 1.0.0, downloaded on 04/01/2022, https://github.com/JinmiaoChenLab/SEDR), which is a dimension reduction method that first uses a deep autoencoder to construct a low-dimensional latent representation of gene expression and then perform spatial domain detection.

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The original public data used in this work can be accessed through the following links: Human dorsolateral prefrontal cortex (DLPFC) data by 10x Visium available at the link: http://spatial.libd.org/spatialLIBD/, with human post-mortem brain snRNA-seq reference data available at Synapse (https://www.synapse.org/#!Synapse:syn18485175); Human squamous cell carcinoma data by Spatial Transcriptomics (ST) is available at GEO accession GSE144240, with the human squamous cell carcinoma scRNA-seq reference data available at GEO accession GSE144236; mouse spermatogenesis data by Slide-seq is available at the link https://www.dropbox.com/s/ygzpj0d0oh67br0/Testis_Slideseq_Data.zip?dl=0, with the mouse testis scRNA-seq reference data available at GEO accession GSE112393; mouse brain (coronal section) Vizgen MERFISH data is available at Vizgen website https://info.vizgen.com/mouse-brain-data, with the mouse brain scRNA-seq reference data is available at http://mousebrain.org/adolescent/; mouse olfactory bulb by Stereo-seq data is available at https://db.cngb.org/stomics/mosta/download/, with the mouse olfactory bulb scRNA-seq data available at GEO accession GSE121891; human breast cancer by 10x Xenium data is available at https://www.10xgenomics.com/products/xenium-in-situ/preview-dataset-human-breast, with the with the human breast cancer scRNA-seq reference data available at GSE accession GSE176078; Detailed description about the data we used in this study is provided in Supplementary Tables 1&2.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No data collection was involved in the present study. We develop our method IRIS and applied IRIS on all publicly available datasets. These publicly available datasets come from 6 studies, representing 6 different popular spatially resolved transcriptomics technologies.
Data exclusions	For each spatial transcriptomics and single cell RNASeq reference dataset, we used all spots or or cells that were detected to have non-zero expression as determined by their original publications. No data were excluded from this study.
Replication	In the nearest neighbor analysis, we calculate p-values by permutation with 1000 times. All attempts at replication were successful.
Randomization	Randomization is not relevant to this study because each dataset from each study is independent with each other
Blinding	Blinding is not relevant to our study because we don't compare any case/control groups.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging